Reasoning about Welfare-Affecting Capabilities in Concurrent Games

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The Motivation

Strategic ability is not only about achieving goals

In multi-agent systems, a coalition's choices may also affect the welfare of other agents.

Classical ATL/CL asks

- Can coalition C ensure φ?
- Can C keep φ true forever?

We also ask

- Can C benefit coalition B?
- Can C harm coalition H?
- Can it do so while ensuring φ?

Who is a potential benefactor, and who is a danger?

Conceptual Ingredients

Harm and benefit

We adopt a consequentialist, non-comparative reading, where good and bad outcomes are represented through agents' preferences:

- harm: bringing about a bad outcome;
- benefit: bringing about a good outcome.

Danger

We follow a modal, non-probabilistic intuition for danger [Fassio, 2025, Garland, 2003, Peschard et al., 2022]:

- a coalition is dangerous if it has the potential to harm;
- potential means existence of a suitable strategy.

strategic capability [Alur et al., 2002, Goranko and van Drimmelen, 2006]

$$+$$

preference over outcomes [Li et al., 2025]

$$\Downarrow$$

welfare-affecting capability

Main Contributions

1. A semantics based on concurrent games enriched with agents' preferences over computations.
2. New ATL and CL modalities for welfare-affecting strategic capabilities:

$$\langle\langle C{:}B{,}H \rangle\rangle\gamma$$

   "C has a strategy that benefits B, harms H, and guarantees γ."
3. Technical results:
   - sound and complete axiomatization for the until-free fragment;
   - polynomial embedding into standard ATL/CL with special atoms marking benefited and harmed agents;
   - tight satisfiability complexity: Exptime-complete for WA$-$ATL, Pspace-complete for WA$-$CL.

Models: Concurrent Games with Preferences

A model is a concurrent game model with regular preferences:

$$M = \left(\mathbb{S}, \mathbb{ACT}, (\mathcal{R}_\delta)_{\delta \in \mathbb{JACT}}, \preceq, \mathcal{V}\right).$$

- $\mathbb{S}$: non-empty set of states.
- $\mathbb{ACT}$: action names; $\mathbb{JACT} = \mathbb{ACT}^n$: action profiles.
- $\mathcal{R}_\delta \subseteq W \times W$: transition relation for action profile δ.
- for each agent i and state s, a regular preference order

$$\preceq_{i,s}$$

  over computations starting at s.
- $\mathcal{V} : \mathbb{S} \longrightarrow 2^{\mathbb{P}}$: valuation function.

Constraints for game structures

Collective choice determinism, independence of choices, and seriality.

Regularity of preferences

The preference order $\preceq_{i,s}$ is regular if it is a total preorder over computations starting at s and:

- stable:
  for all computations λ, λ' starting at s, if $\lambda(1) = \lambda'(1)$ and $\lambda \preceq_{i,s} \lambda'$, then $\lambda_{\geq 1} \preceq_{i,s'} \lambda'_{\geq 1}$, where $s' = \lambda(1)$;
- well-founded:
  the worst computation exists;
- conversely well-founded:
  the best computation exists;
- non-flat:
  there is λ, λ' starting at s such that $\lambda \prec_{i,s} \lambda'$.

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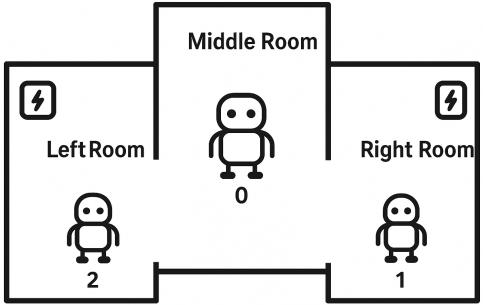
Running Example: Robots and Charging Rooms



- Robot 0 starts in the middle room.
- It wants to reach a charging station on the left or right.
- Robots 1 and 2 guard the right and left passages.
- Each guard can block or free its passage.

Preference profile

Robot 0 prefers computations where it eventually reaches a charging station.
Robots 1 and 2 prefer computations where robot 0 never reaches one.

Strong Benefit and Strong Harm

Let $\mathcal{O}_{\mathsf{F}}(s, \mathcal{F}_C)$ be the computations compatible with strategy $\mathcal{F}_C$ from s.

Strongly beneficial for B

$$\mathcal{F}_C \in StrSB^C_{\mathsf{M}}(s, B) \quad \text{iff} \quad \mathcal{O}_{\mathsf{F}}(s, \mathcal{F}_C) \subseteq Best_{i,s} \text{ for every } i \in B.$$

Strongly harmful for H

$$\mathcal{F}_C \in StrSH^C_{\mathsf{M}}(s, H) \quad \text{iff} \quad \mathcal{O}_{\mathsf{F}}(s, \mathcal{F}_C) \subseteq Worst_{i,s} \text{ for every } i \in H.$$

Strong benefit/harm means guaranteeing best/worst outcomes for the affected coalition.

Fact

The coalition of robots $\{1, 2\}$ has a strategy that is harmful to robot 0 and beneficial to the guarding coalition.

The Language WA−ATL

$$
\begin{aligned}
\varphi, \psi &::= p \mid \neg\varphi \mid \varphi \wedge \psi \mid \langle\langle C{:}B,H \rangle\rangle\gamma \\
\gamma &::= \mathsf{X}\varphi \mid \mathsf{G}\varphi \mid \varphi \mathsf{U} \psi
\end{aligned}
$$

Key modality

$$\langle\langle C{:}B,H \rangle\rangle\gamma$$

Coalition C has a strategy which is beneficial for B, harmful to H, and guarantees temporal property γ.

Standard ATL is recovered as the special case:

$$\langle\langle C \rangle\rangle\gamma \equiv \langle\langle C{:}\emptyset,\emptyset \rangle\rangle\gamma.$$

Semantics of the New Modality

$$(\mathsf{M}, s) \models \langle\!\langle C{:}B{,}H \rangle\!\rangle \gamma$$

iff there exists a strategy

$$\mathcal{F}_C \in StrSB^C_{\mathsf{M}}(s, B) \cap StrSH^C_{\mathsf{M}}(s, H)$$

such that every generated computation satisfies γ:

$$\forall \lambda \in \mathcal{O}_{\mathsf{F}}(s, \mathcal{F}_C), \quad (\mathsf{M}, \lambda) \models \gamma.$$

Three roles in one operator

- C: the acting coalition;
- B: the coalition benefited by the strategy;
- H: the coalition harmed by the strategy.

Potential Benefactor and Danger

The language gives compact definitions of the target notions.

Potential benefactor

$$\mathrm{Benef}(C, B) \;\equiv\; \langle\!\langle C{:}B{,}\emptyset \rangle\!\rangle \mathsf{X}\top.$$

Coalition C has a strategy beneficial for B.

Danger

$$\mathrm{Danger}(C, H) \;\equiv\; \langle\!\langle C{:}\emptyset{,}H \rangle\!\rangle \mathsf{X}\top.$$

Coalition C has a strategy harmful to H.

In the robot model:

$$(\mathrm{M}_{rs}, s_0) \models \mathrm{Benef}(\{1,2\},\{1,2\}) \wedge \mathrm{Danger}(\{1,2\},\{0\}).$$

We axiomatize the until-free, or safety, fragment:

$$\varphi ::= p \mid \neg\varphi \mid \varphi \wedge \varphi \mid \langle\!\langle C{:}B,H \rangle\!\rangle \mathsf{X}\varphi \mid \langle\!\langle C{:}B,H \rangle\!\rangle \mathsf{G}\varphi.$$

Inference rules

$$\frac{\varphi, \quad \varphi \rightarrow \psi}{\psi} \ (\mathrm{MP}) \qquad \frac{\varphi \leftrightarrow \psi}{\langle\!\langle C{:}B,H \rangle\!\rangle \mathsf{X}\varphi \leftrightarrow \langle\!\langle C{:}B,H \rangle\!\rangle \mathsf{X}\psi} \ (\mathrm{RE}) \qquad \frac{\varphi}{\langle\!\langle C \rangle\!\rangle \mathsf{G}\varphi} \ (\mathrm{Gen})$$

Axiomatization

Axiom schemata

All tautologies of propositional logic	(TAUT)
$\neg\langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{X}\bot$	(NAAA)
$\langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{X}(\varphi \wedge \psi) \to \langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{X}\varphi$	(MG)
$\langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{X}\varphi \to \langle\!\langle C'{:}B',H'\rangle\!\rangle \mathsf{X}\varphi$ for $C \subseteq C'$, $B \supseteq B'$, $H \supseteq H'$	(MC)
$\langle\!\langle \emptyset\rangle\!\rangle \mathsf{X}\top$	(NEI)
$(\langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{X}\varphi \wedge \langle\!\langle C'{:}B',H'\rangle\!\rangle \mathsf{X}\psi) \to \langle\!\langle C \cup C'{:}B \cup B',H \cup H'\rangle\!\rangle \mathsf{X}(\varphi \wedge \psi)$ for $C \cap C' = \emptyset$	(Sup)
$\langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{X}(\varphi \vee \psi) \to (\langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{X}\varphi \vee \langle\!\langle \mathbb{AGT}{:}B,H\rangle\!\rangle \mathsf{X}\psi)$	(Cro)
$\langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{G}\varphi \leftrightarrow \langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{X}\big(\varphi \wedge \langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{G}\varphi\big)$	(FP_{G})
$\langle\!\langle \emptyset\rangle\!\rangle \mathsf{G}\big(\psi \to \langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{X}(\varphi \wedge \psi)\big) \to \langle\!\langle \emptyset\rangle\!\rangle \mathsf{G}\big(\psi \to \langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{G}\varphi\big)$	($\mathrm{GFP}_{\mathsf{G}}$)
$\langle\!\langle \mathbb{AGT}{:}\emptyset,\{a\}\rangle\!\rangle \mathsf{X}\top$	(WFP)
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$\langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{X}(\varphi \lor \psi) \to (\langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{X}\varphi \lor \langle\!\langle \mathbb{AGT}{:}B,H\rangle\!\rangle \mathsf{X}\psi)$	(Cro)
$\langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{G}\varphi \leftrightarrow \langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{X}(\varphi \land \langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{G}\varphi)$	($\mathrm{FP_G}$)
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Axiomatization

Theorem

For every $\varphi \in \mathcal{L}_{\mathsf{WA-ATL-U}}$:

$$\varphi \text{ is valid over CGMRPs} \quad \text{iff} \quad \vdash \varphi.$$

- The proof uses canonical models built under a formula context.
- The states in the canonical models are consist of (*finite*) maximally consistent sets of formulas in the formula context, and the labels of benefited and harmed groups.
- Stability of preferences keeps those labels coherent over time.

Tree-Like Model Property

Theorem

For every $\varphi \in \mathcal{L}_{\mathsf{WA-ATL}}$:

$$\varphi \text{ is satisfiable} \quad \text{iff} \quad \varphi \text{ is satisfiable on a pointed tree-like CGMRP.}$$

- Tree-like models have a unique root, unique predecessors, and no cycles.
- This lets us localize welfare labels along generated computations.

Embedding into Standard ATL

Extend the atomic vocabulary with:

$$\heartsuit_i \quad \text{and} \quad \spadesuit_i$$

read as "agent i is benefited" and "agent i is harmed".

Translation:

$$\begin{aligned}
tr(\langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{X}\varphi) &= \langle\!\langle C\rangle\!\rangle \mathsf{X}\big(\heartsuit B \wedge \spadesuit H \wedge tr(\varphi)\big),\\
tr(\langle\!\langle C{:}B,H\rangle\!\rangle \mathsf{X}\varphi) &= \langle\!\langle C\rangle\!\rangle \mathsf{G}\big(\heartsuit B \wedge \spadesuit H \wedge tr(\varphi)\big),\\
tr(\langle\!\langle C{:}B,H\rangle\!\rangle (\varphi \mathsf{U} \psi)) &= \langle\!\langle C\rangle\!\rangle \big((\heartsuit B \wedge \spadesuit H \wedge tr(\varphi))\mathsf{U}(\heartsuit B \wedge \spadesuit H \wedge tr(\psi))\big),
\end{aligned}$$

where

$$\heartsuit B = \bigwedge_{i\in B} \heartsuit_i, \qquad \spadesuit H = \bigwedge_{i\in H} \spadesuit_i.$$

Embedding Theorem and Complexity

Polynomial embedding

Satisfiability for WA−ATL reduces polynomially to satisfiability for standard ATL, with additional constraints ensuring:

best labels exist, worst labels exist, and no state is both for the same agent.

Complexity

Language	Target	Satisfiability
WA−ATL	ATL	Exptime-complete
WA−CL	CL	Pspace-complete

The bounds are tight because ATL and CL are special cases.

Summary and Outlook

- Strategic agency can be welfare-affecting: agents can benefit or endanger others through their strategies.
- WA−ATL internalizes this idea in the object language:

$$\langle\!\langle C{:}B,H \rangle\!\rangle \gamma.$$

- Potential benefactor and danger become definable notions.
- The logic remains technically well-behaved:
  - complete axiomatization for the safety fragment;
  - polynomial embedding into standard ATL/CL with welfare-label atoms;
  - tight complexity bounds.

Next steps

Full axiomatization with until operators; probabilistic risk-based danger.

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Thank you

Questions?